

Project Design Phase- II

Data Flow Diagram & User Stories

Date	28 June 2026
Roll no	23A51A1259
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

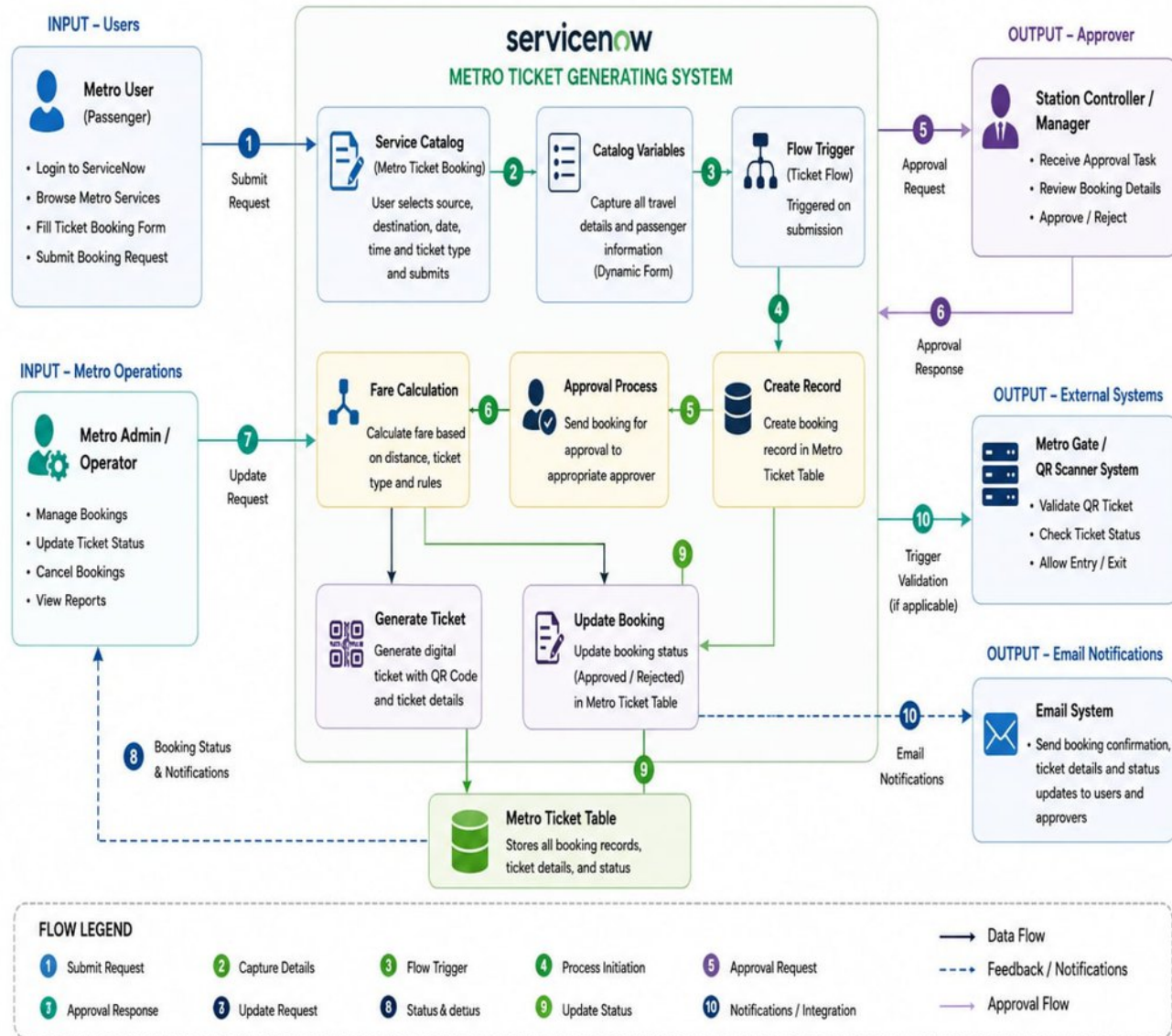
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows through the Metro Ticket Generating System in ServiceNow. It shows how users interact with the system, how booking information is processed, where data is stored, and how the final ticket is generated.

In the Metro Ticket Generating System, the passenger enters journey details such as the source station, destination station, and travel information through the ServiceNow Service Catalog. The system validates the entered details, automatically calculates the ticket fare, and stores the booking information in the database. The Flow Designer then processes the request, generates a unique QR- code- based digital ticket, and sends a confirmation notification to the passenger. The generated ticket and booking details are securely maintained in the system for future reference and reporting.

The Data Flow Diagram helps visualize the movement of data between the passenger, ServiceNow application, database, and notification system. It provides a clear understanding of how information is collected, processed, stored, and delivered, ensuring an efficient, secure, and automated metro ticket booking process.

DATA FLOW DIAGRAM – Metro Ticket Generating System in ServiceNow



USM- 1: As a metro user, I can book a metro ticket through the Service Catalog.

USM- 2: As a metro user, I can select the source station, destination station, travel date, and number of passengers before submitting the booking.

USM- 3: As a metro user, I can view the automatically calculated ticket fare before confirming my booking.

USM- 4: As a metro user, I can view dynamic form fields based on the selected journey details or ticket type.

USM- 5: As a metro user, I can receive a QR- code- based digital ticket after successful booking.

USM- 6: As a metro user, I can track the status of my ticket booking.

USM- 7: As a metro user, I can receive email notifications containing booking confirmation and ticket details.

USM- 8: As a metro operator, I can view all submitted ticket booking requests.

USM- 9: As a metro operator, I can update the booking status, cancel bookings if required, and manage passenger requests.

USM- 10: As a metro operator, I can verify and validate QR- code tickets during passenger entry.

USM- 11: As a station manager, I can approve or reject special ticket booking requests (if approval is required).

USM- 12: As a station manager, I can provide comments while approving or rejecting a booking request.

USM- 13: As an administrator, I can create and manage Metro Ticket Booking catalog items.

USM- 14: As an administrator, I can configure catalog variables and variable sets for ticket booking forms.

USM- 15: As an administrator, I can configure Catalog UI Policies to display dynamic booking fields.

USM- 16: As an administrator, I can automate ticket booking, fare calculation, QR code generation, and notifications using Flow Designer.

USM- 17: As an administrator, I can create and maintain the Metro Ticket Booking database table.

USM- 18: As an administrator, I can generate booking reports and monitor metro ticket booking performance.